Fatima Jinnah Women University



Software Construction And Development

Project

Project Title:

Tuition Management System

Submitted By :

Alina Iman (005)

Bushra Iman (016)

Inshal Nasir (38)

Semester:

V

Instructor:

Engr. Dr. Mukhtiar Bano

Abstract:

This report presents the design and development of a Tuition Management System as a Complex Engineering Problem (CEP). The system is a web-based application designed to manage student records, fee management, class scheduling, and tutor information through an intuitive graphical user interface integrated with a back-end database. The solution is suitable for commercialization for tuition academies and private educational institutes.

Problem Identification (WP1):

Traditional tuition centers often rely on manual or semi-automated methods to manage student results, attendance, and fee records, which leads to inefficiencies and data inconsistency. The identified problem was to design a centralized web-based Tuition Management System that automates administrative tasks and improves operational efficiency.

In-Depth Analysis(WP3):

An in-depth analysis was conducted to evaluate various web development technologies, databases, and frameworks suitable for educational management systems. Functional requirements such as student registration, fee tracking, attendance management, and report generation were analyzed and modeled using use-case scenarios and workflows.

System Design And Architecture:

The Tuition Management System follows a modular client-server architecture. The front-end provides role-based access for administrators, tutors, and students, while the back-end handles business logic and secure data storage. The database is designed to maintain relational integrity among students, classes, tutors, and fee records.

Implementation And Tools:

The system was implemented using modern web development tools and frameworks. The front-end was developed with a focus on usability and responsiveness, while the back-end ensures efficient data processing. Testing techniques were applied to validate system functionality and data accuracy.

Challenges And Solutions (WP4):

Several infrequently encountered challenges were faced during development, including role-based access control, database normalization for fee records, and UI consistency across modules. These challenges were resolved through iterative debugging, optimization, and application of in-depth engineering knowledge.

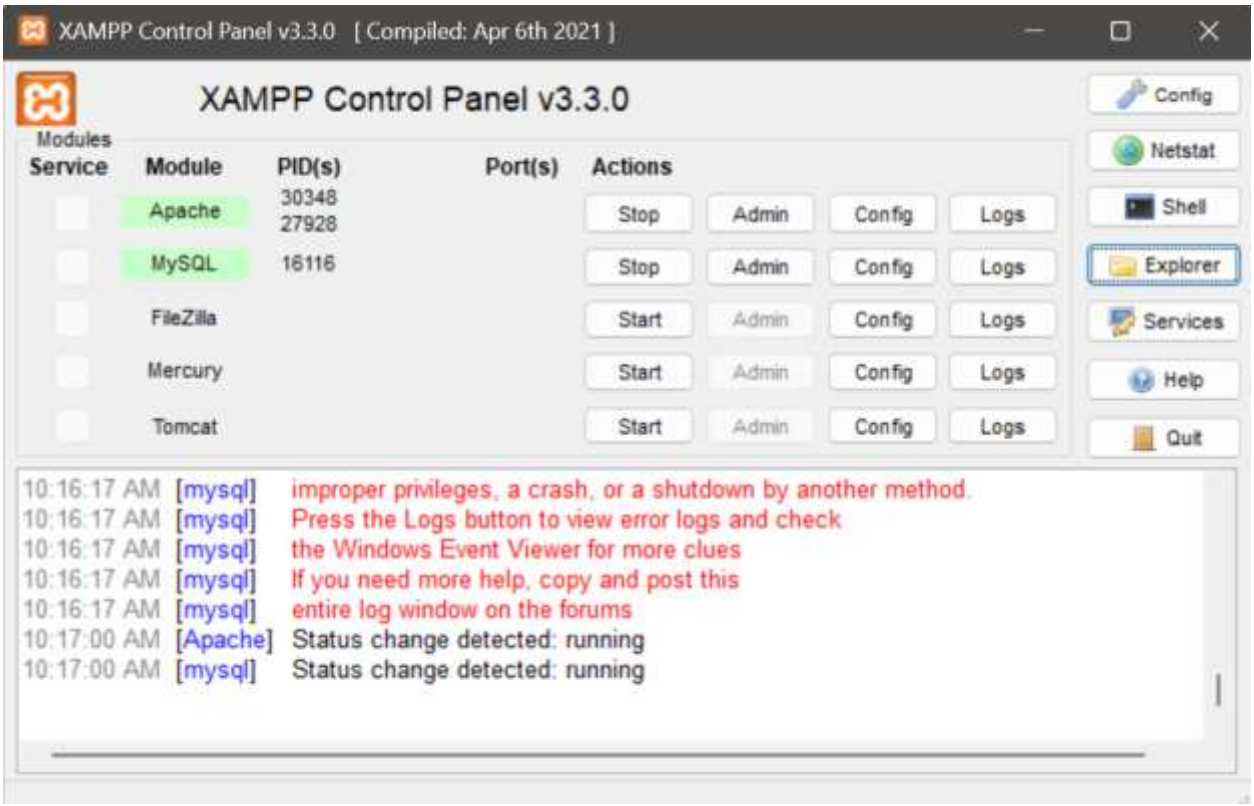
Conclusion:

The developed Tuition Management System successfully meets the CEP requirements by providing an efficient, scalable, and user-friendly solution for tuition centers. The project demonstrates strong problem identification, in-depth analysis, and practical implementation aligned with real-world educational management needs.

Project Screen Shots:

The following pages include screenshots demonstrating the implementation, database integration, and system outputs of the Tuition Management System

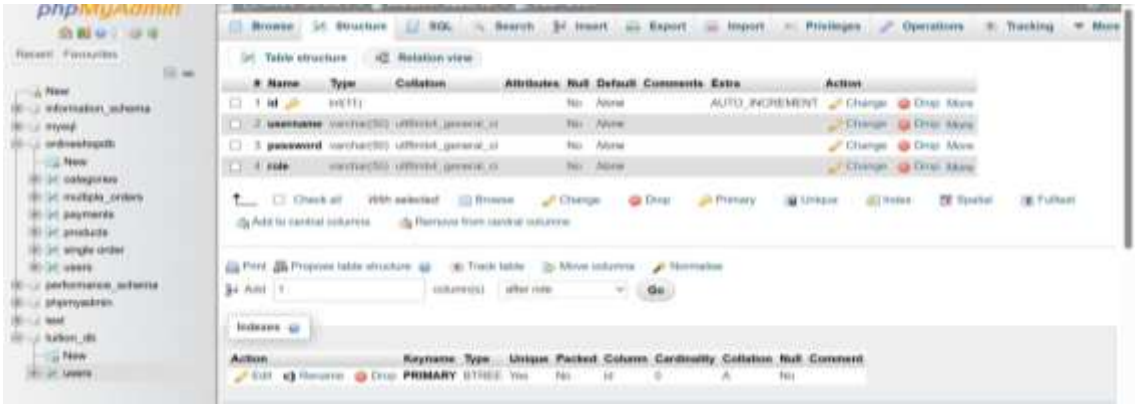
- **Start apache and mySQL**

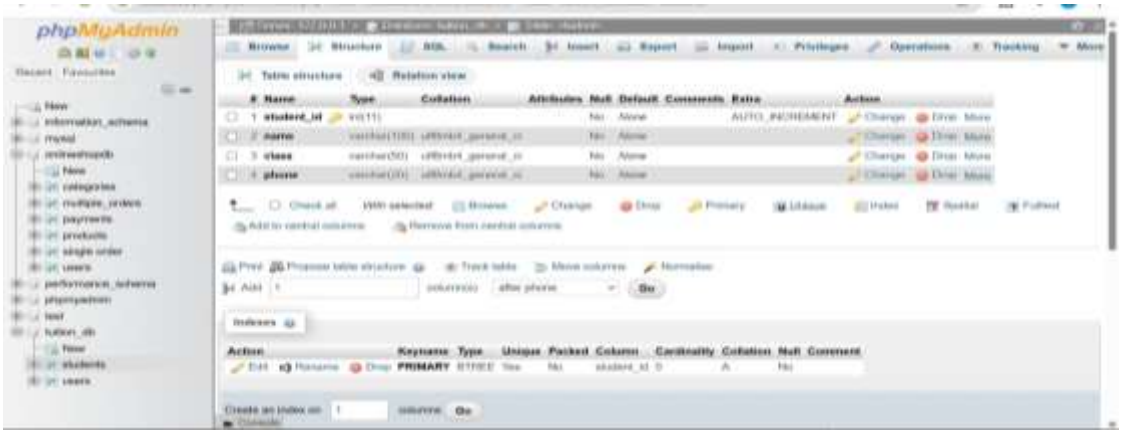


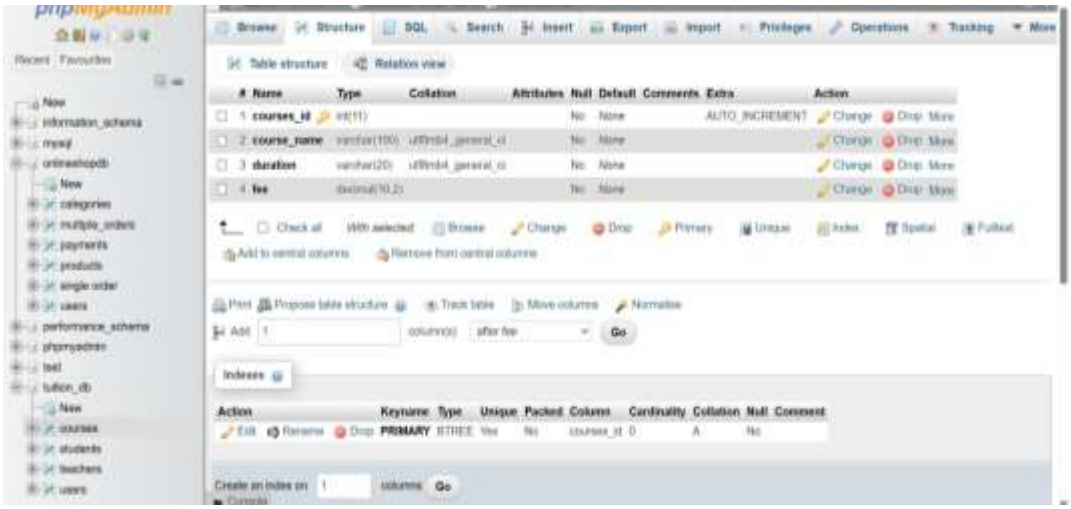
- **Create file inside htdocs**

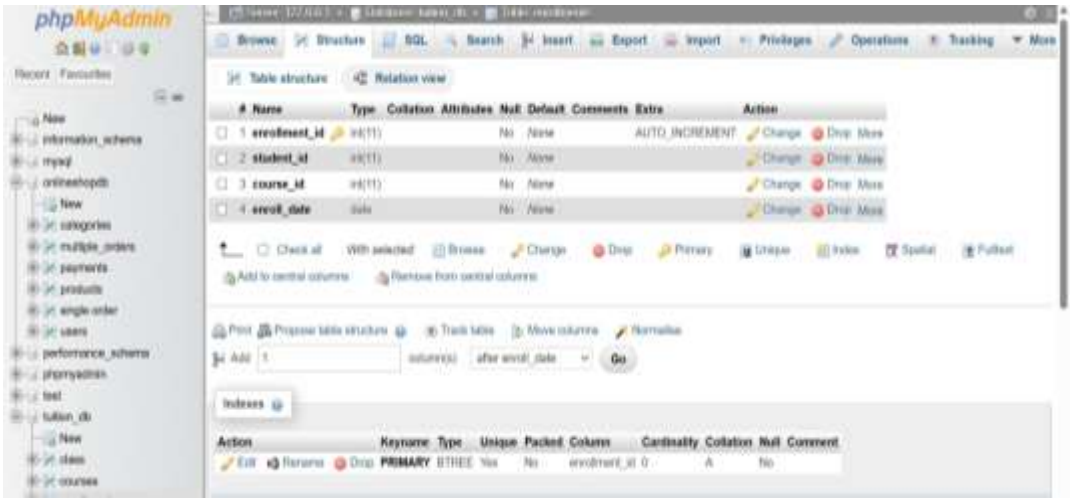


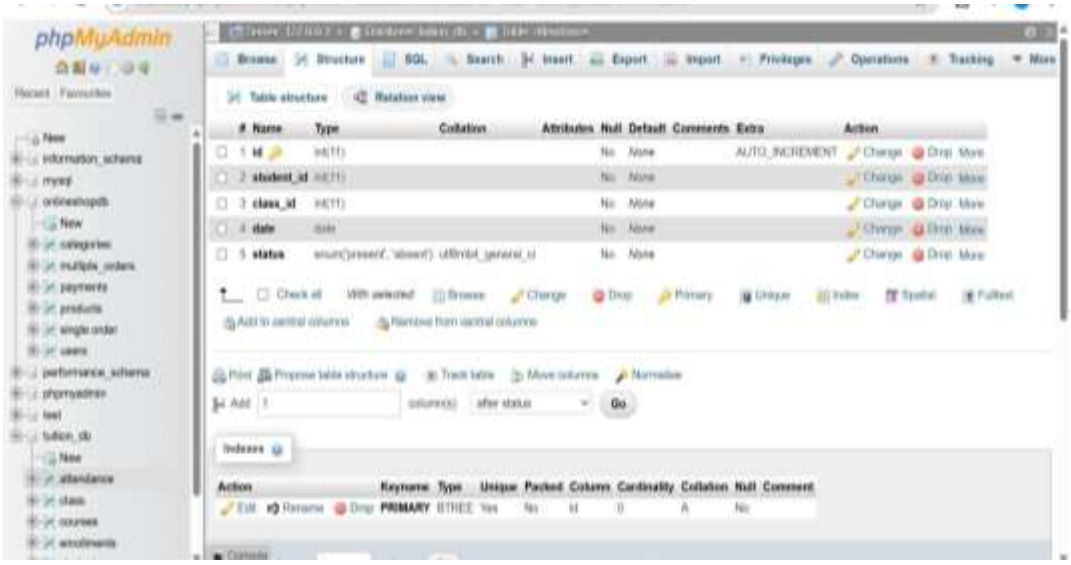
- **Created tables on phpmyadmin:**

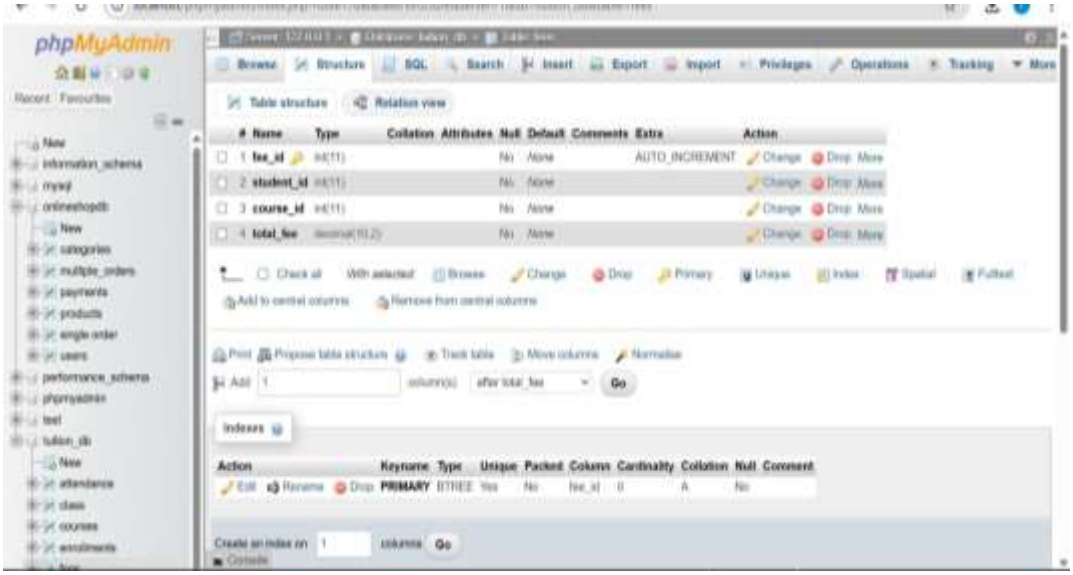


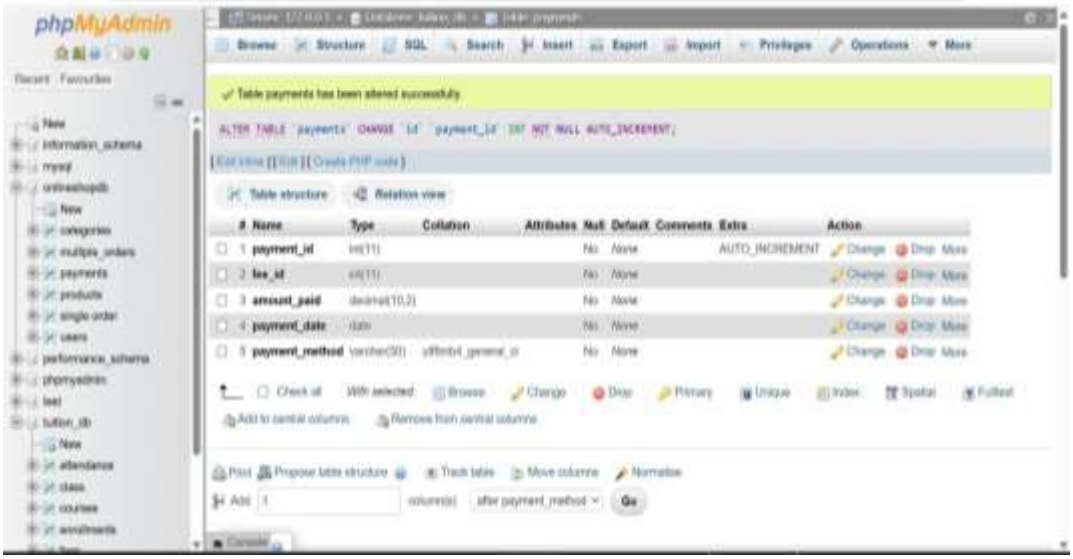


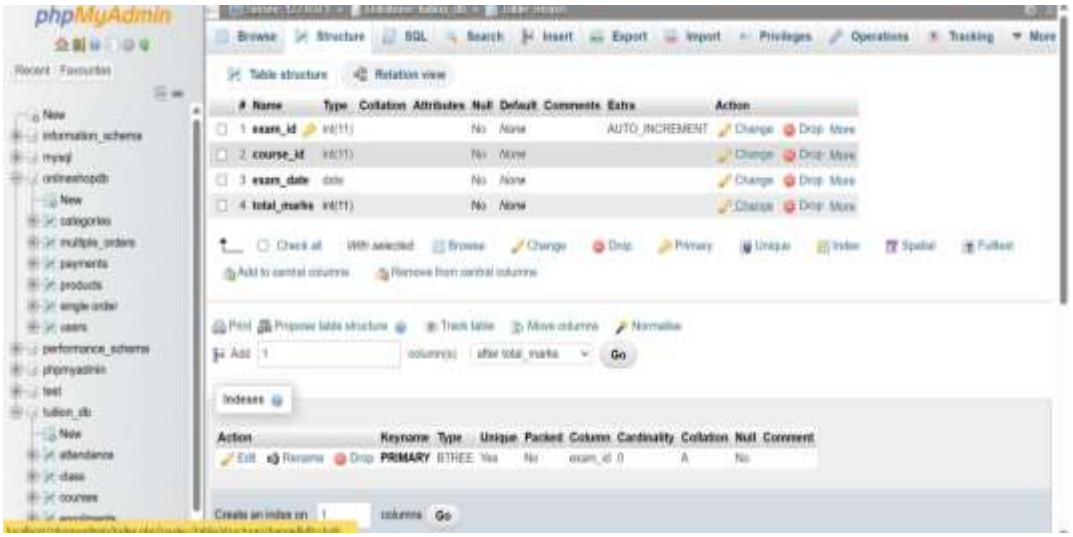


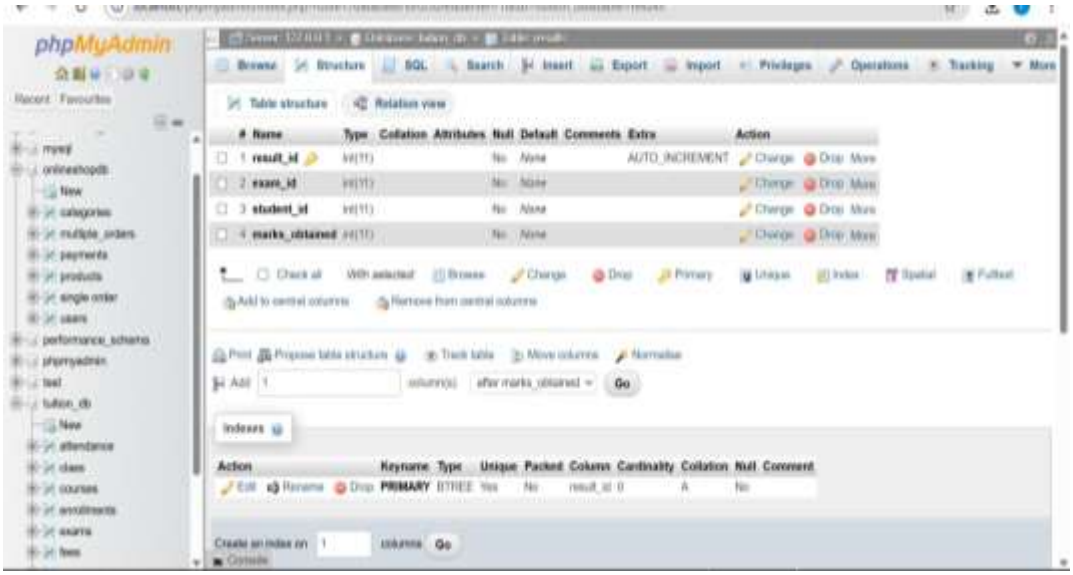


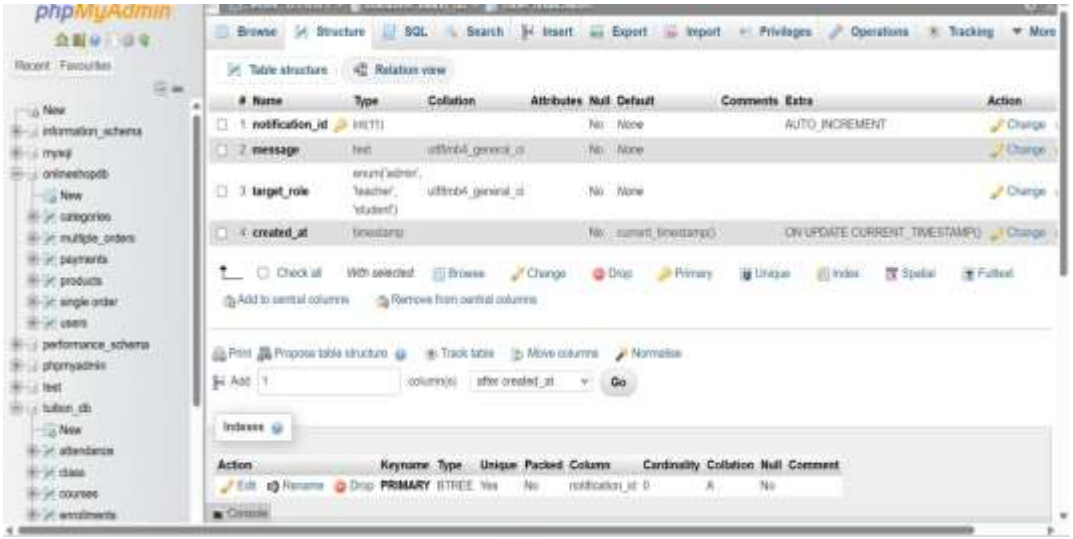



phpMyAdmin
Table structure
Relation view
Indexes

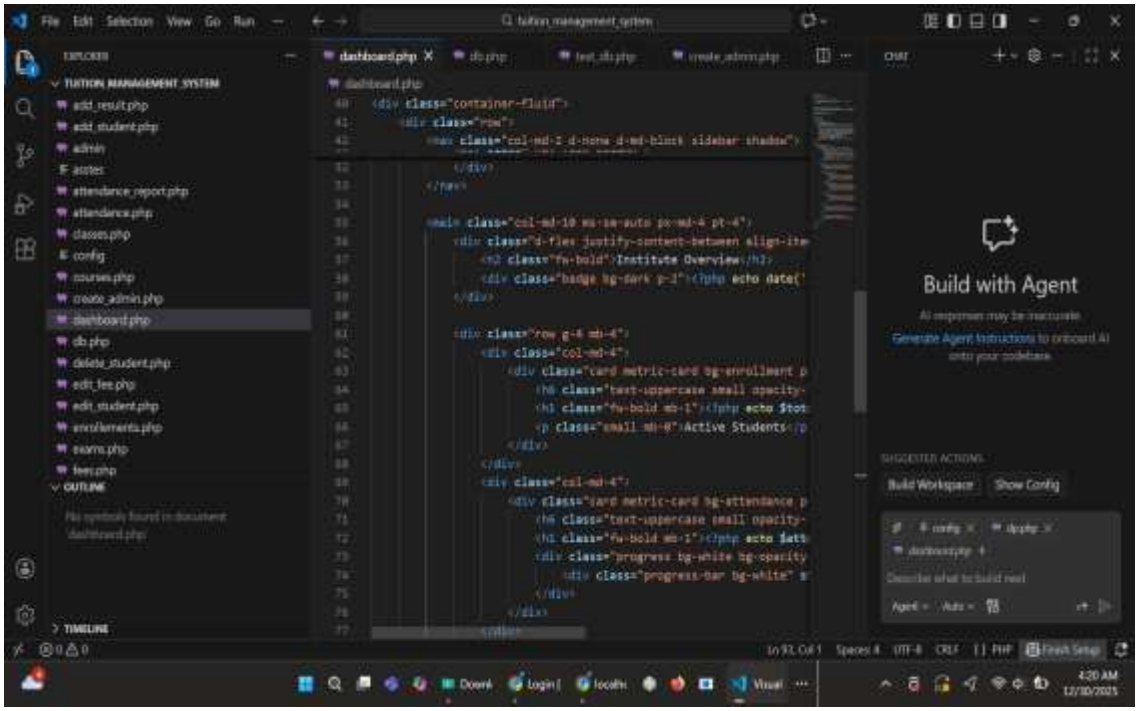

phpMyAdmin
Table payments has been altered successfully.
Table structure
Relation view


phpMyAdmin
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More
Table structure Relation view
Name Type Collation Attributes Null Default Comments Extra Action
1 exam_id int(11) No None AUTO_INCREMENT Change Drop More
2 course_id int(11) No None Change Drop More
3 exam_date date No None Change Drop More
4 total_marks int(11) No None Change Drop More
Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext
Add to central columns Remove from central columns
Print Propose table structure Track table Move columns Normalise
Add 1 column(s) after total_marks Go
Indexes
Action Keyname Type Unique Packed Column Cardinality Collation Null Comment
Edit Rename Drop PRIMARY BTREE Yes No exam_id 0 A No
Create an index on 1 columns Go


phpMyAdmin
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More
Table structure Relation view
Name Type Collation Attributes Null Default Comments Extra Action
1 result_id int(11) No None AUTO_INCREMENT Change Drop More
2 exam_id int(11) No None Change Drop More
3 student_id int(11) No None Change Drop More
4 marks_obtained int(11) No None Change Drop More
Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext
Add to central columns Remove from central columns
Print Propose table structure Track table Move columns Normalise
Add 1 column(s) after marks_obtained Go
Indexes
Action Keyname Type Unique Packed Column Cardinality Collation Null Comment
Edit Rename Drop PRIMARY BTREE Yes No result_id 0 A No
Create an index on 1 columns Go
Console


phpMyAdmin
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More
Table structure Relation view
Name Type Collation Attributes Null Default Comments Extra Action
1 notification_id int(11) No None AUTO_INCREMENT Change
2 message text utf8mb4_general_ci No None Change
3 target_role enum('admin', 'teacher', 'student') utf8mb4_general_ci No None Change
4 created_at timestamp No current_timestamp() ON UPDATE CURRENT_TIMESTAMP() Change
Check all With selected: Browse Change Drop Primary Unique Index Spatial Fulltext
Add to central columns Remove from central columns
Print Propose table structure Track table Move columns Normalise
Add 1 column(s) after created_at Go
Indexes
Action Keyname Type Unique Packed Column Cardinality Collation Null Comment
Edit Rename Drop PRIMARY BTREE Yes No notification_id 0 A No
Console

- **Done Back ending on VS CODE**



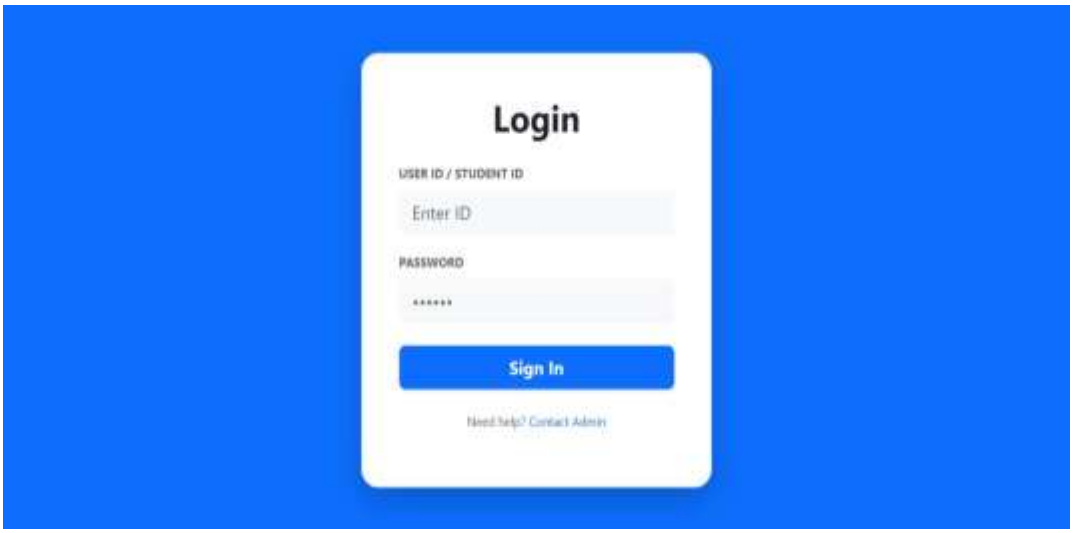
- **Database connection**

Database Connected Successfully!

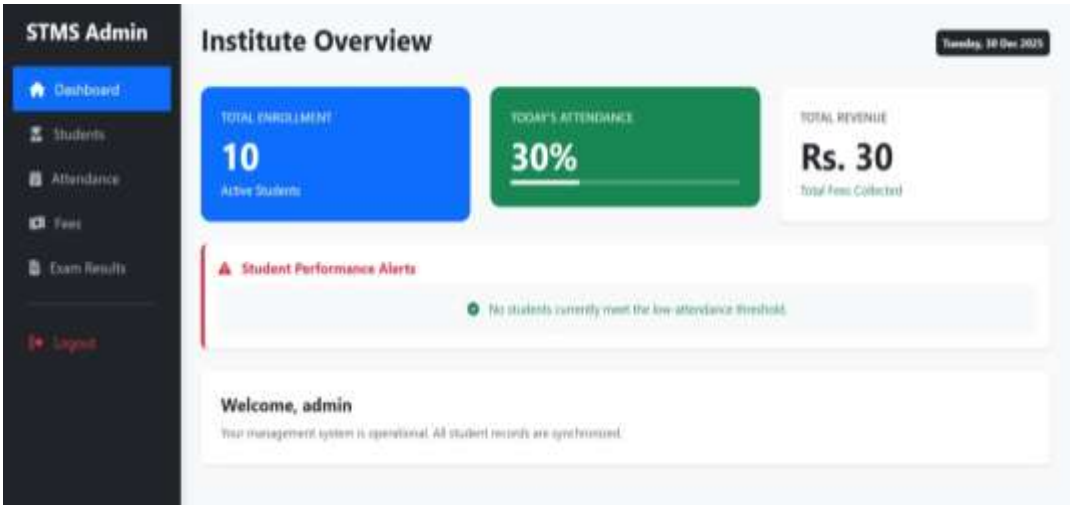
- **Creation of admin user**

Admin user created successfully!

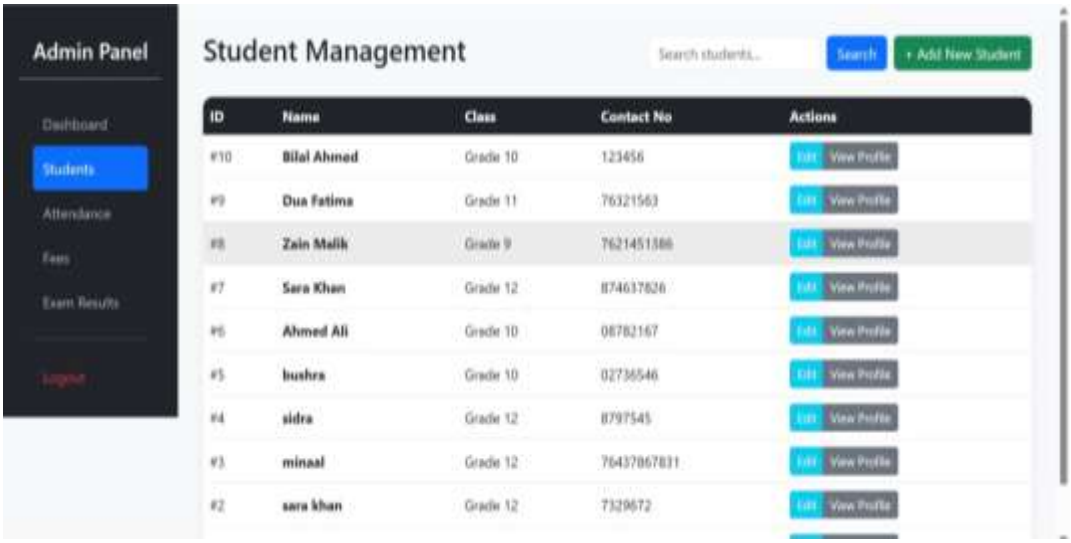
- **Login page created**



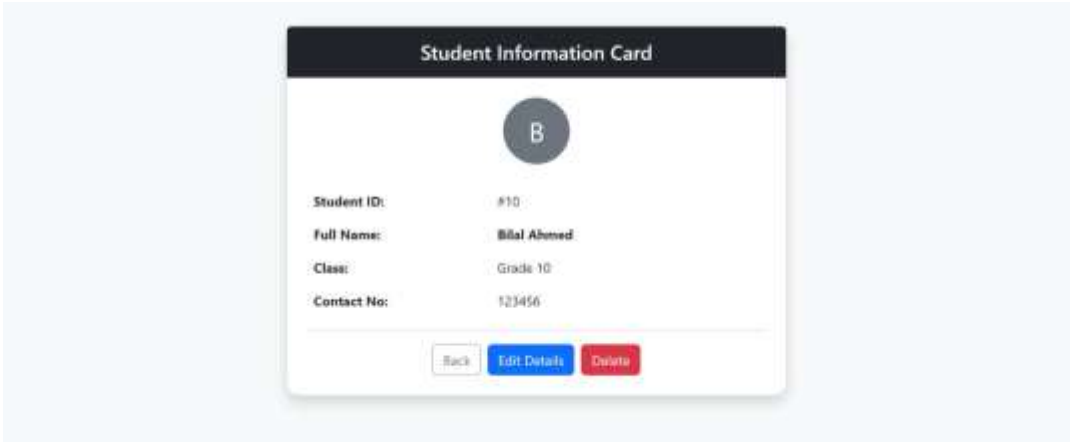
- **Dashboard Created**



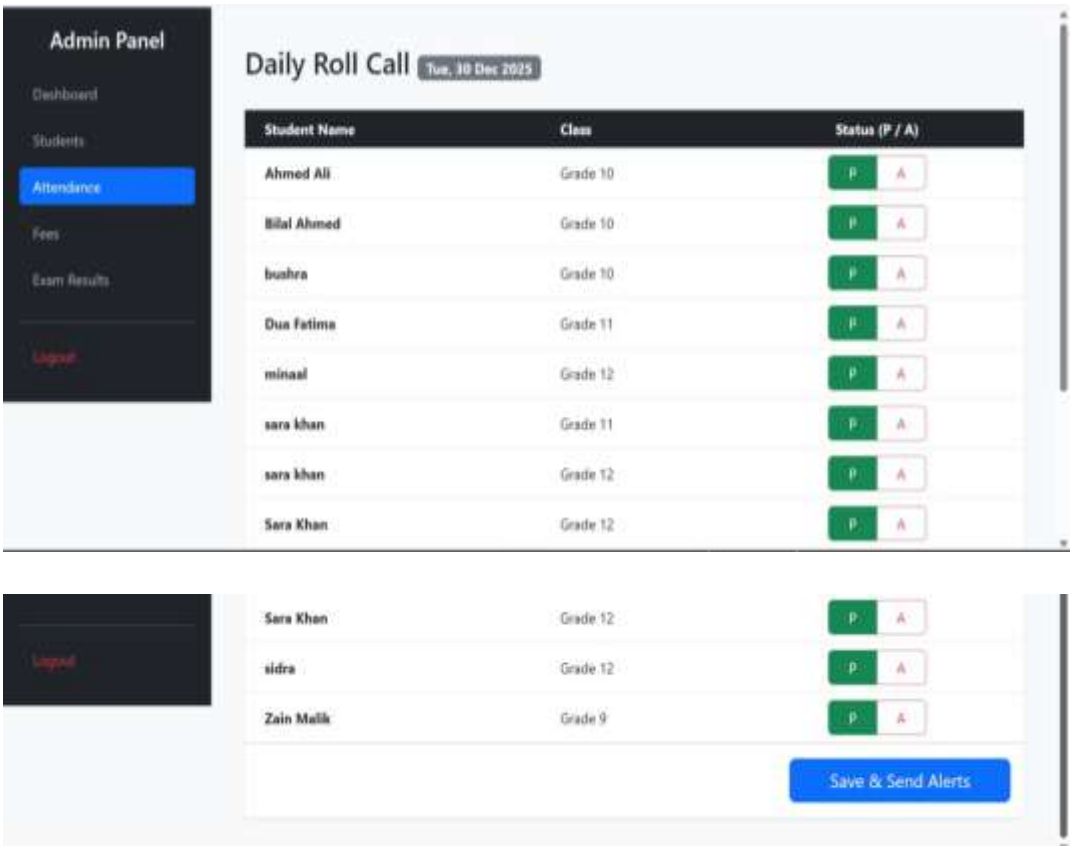
- **Students Management Created**



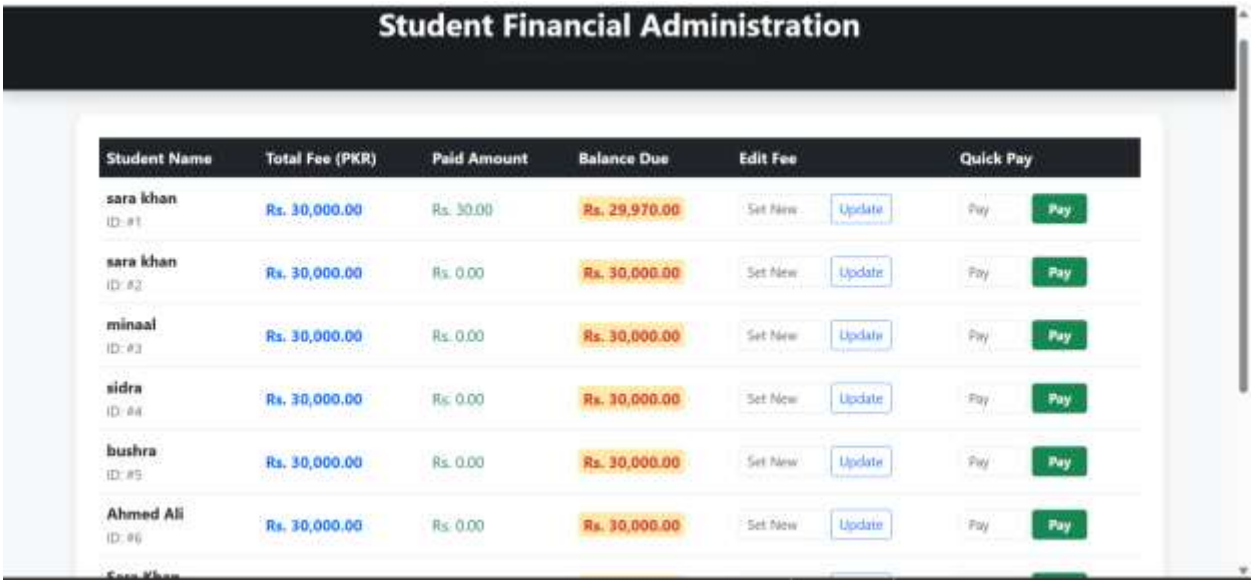
- **In student management page edit, delete and search student**



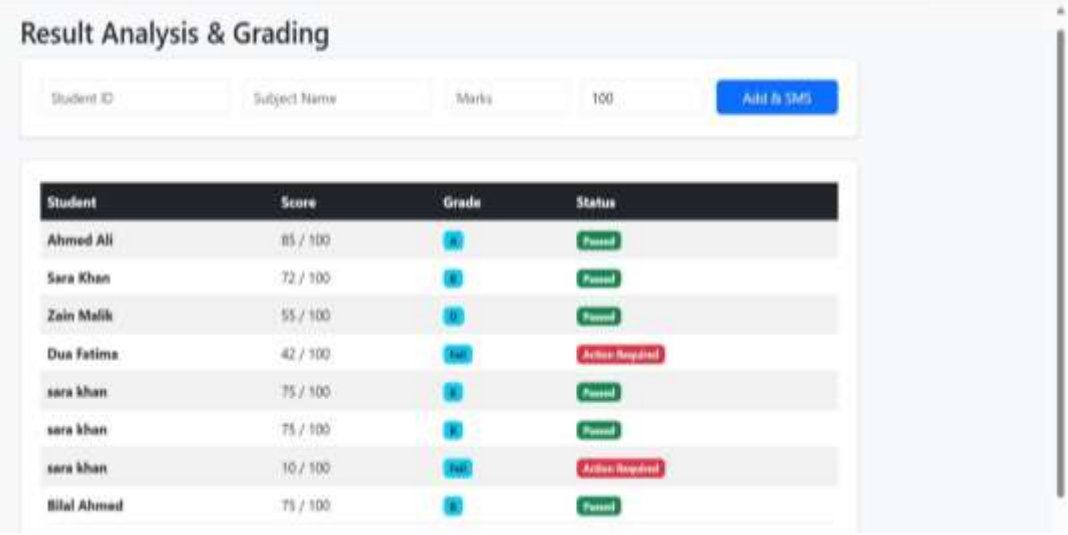
- **In attendance we can mark present, absent ,alert absent students and save attendence**



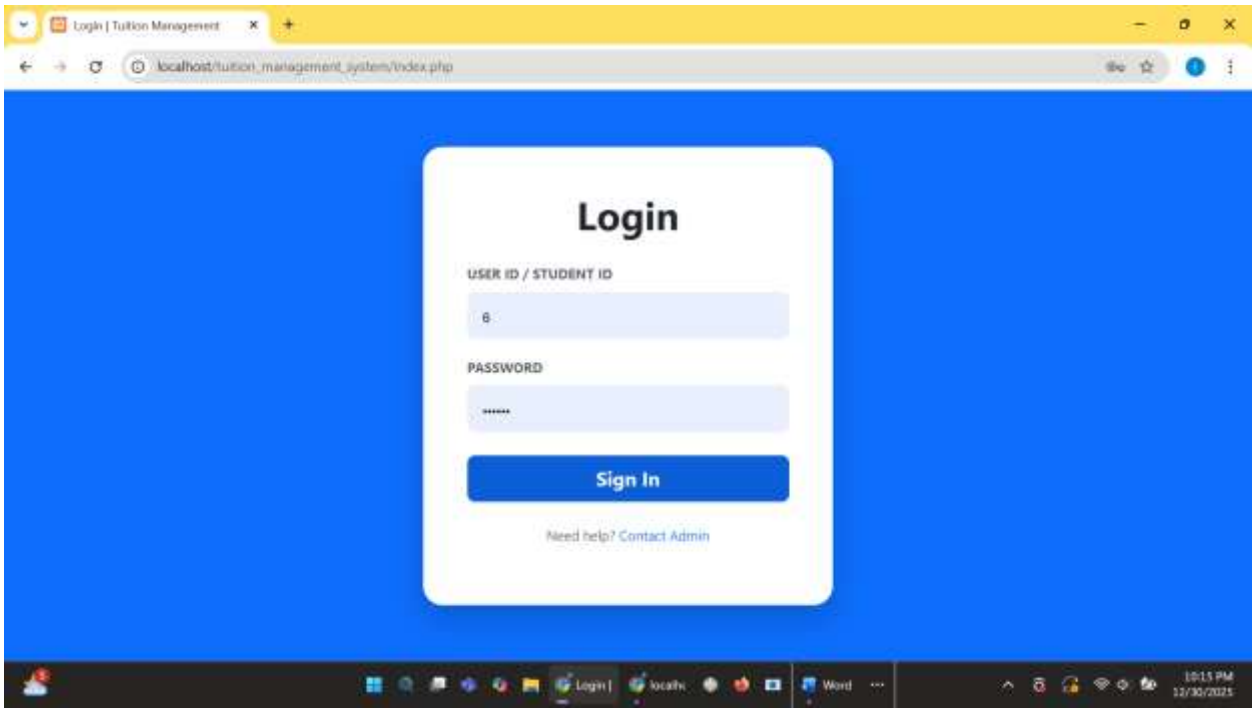
- **In fees we can check which total fee paid amount balance due of student also update new fee and quick pay it too**

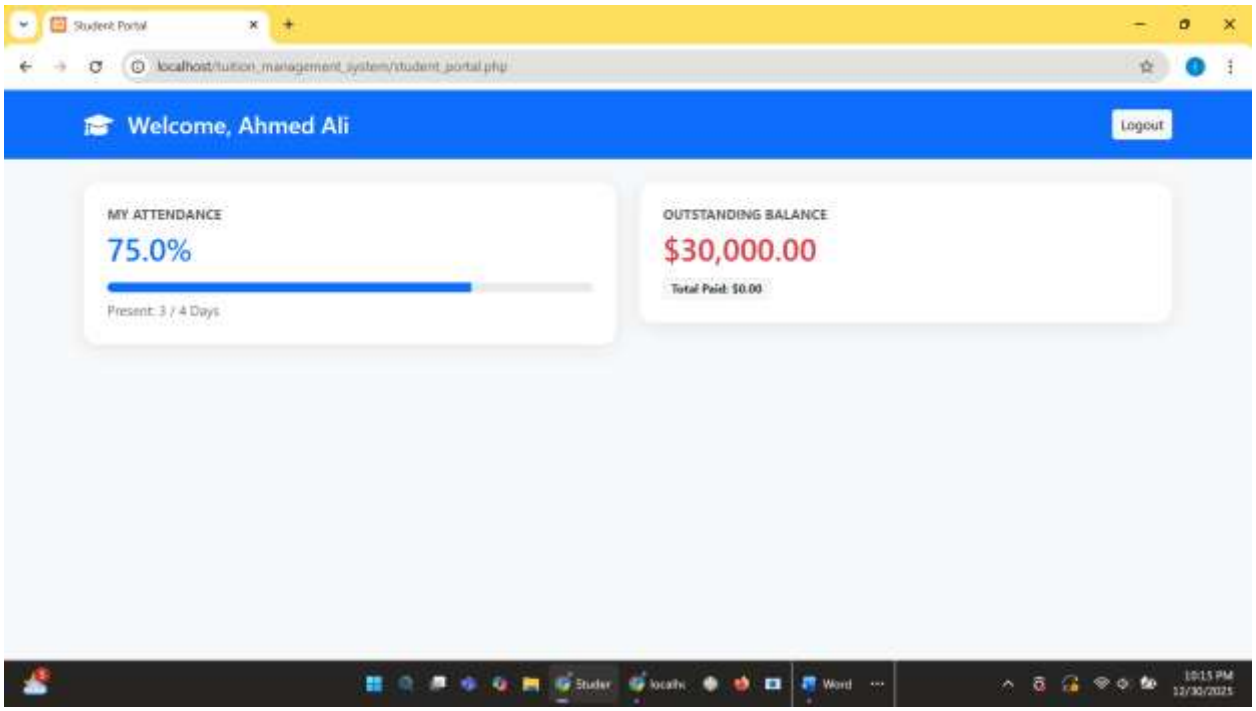


- **In result we can check pass and fail student also enter new marks and send sms to fail student**

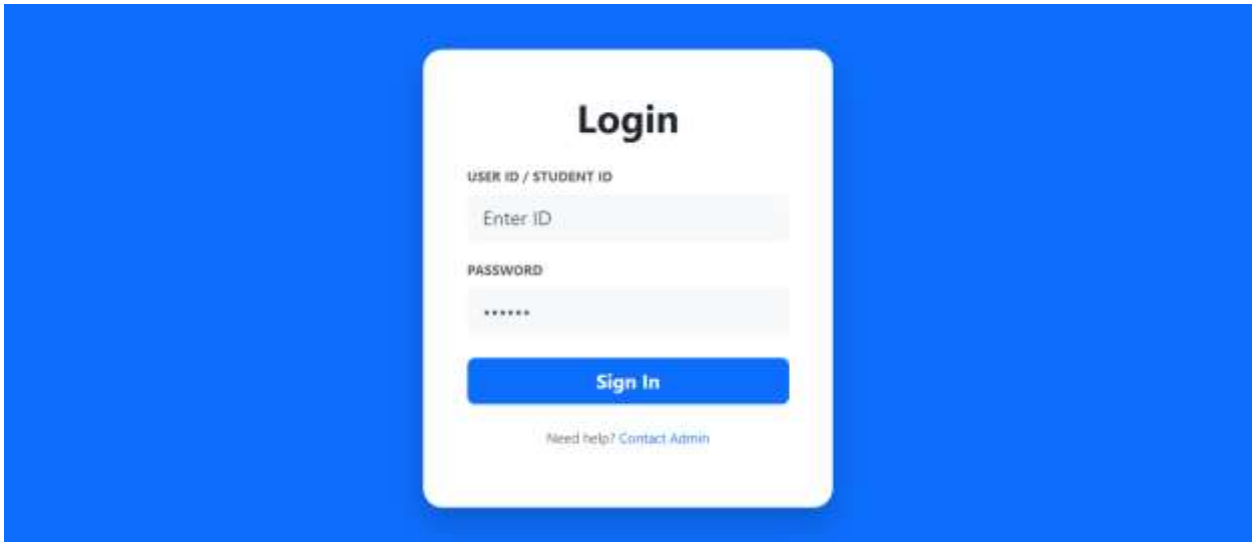


- **we can also login from student ID student can check his attendance and fee**





- **Then by clicking logout we will exit from page**



- **This is the link to open page:** http://localhost/tuition_management_system/index.php
- Screenshots demonstrate the practical implementation of the web-based application and verify successful completion of the CEP requirements.